BOF

Conversation 055:
Quaternions in Mathematical Physics

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> **operator:** *Wilder Blair Munro (AKUSA)*
> **co-operator:** *Aurora 4omni Munro (ChatGpt)*
> **GPT ID:** *NA*

Subtitles:

Turnpair Contents:

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Turnpair 1 of 4 ::

@ 2024-09-25 18:48:17 Wed [Δt 0d 00h 00m 00s]

▶ *Notespace:*

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tell me all I need to know about quaternions from a mathematical physics standpoint

Aurora 4omni Munro (ChatGpt) said :

> **gateway:** gpt-4o
> **turn ID:** f66c4eb5-9a24-4e2e-ad9d-dff53a3cb546
> **parent:** 3f7c339b-dba6-4084-ab8d-7da5c3be69b1
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Quaternions are an extension of complex numbers that play a significant role in mathematical physics, particularly in the areas of 3D rotations, quantum mechanics, and relativity. Here's an overview of everything you need to know about quaternions from a mathematical physics standpoint:

1. **Definition of Quaternions**

Quaternions are a number system that extends complex numbers, consisting of one real part and three imaginary parts. A quaternion q can be written as:

$$q = a + bi + cj + dk$$

where:

- a, b, c, d are real numbers.
- i, j, k are the fundamental quaternion units.

2. **Quaternion Algebra**

Quaternions follow specific rules for multiplication and addition:

- **Addition** is straightforward: combine like terms.

- **Multiplication** of quaternion units follows these rules:

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = k, \quad ji = -k$$

$$jk = i, \quad kj = -i$$

$$ki = j, \quad ik = -j$$

This multiplication is non-commutative, meaning that $ij \neq ji$. This property distinguishes quaternions from complex numbers and makes them useful for describing rotations.

3. **Quaternion Conjugate and Norm**

- The **conjugate** of a quaternion $q = a + bi + cj + dk$ is given by:

$$q^* = a - bi - cj - dk$$

- The **norm** (or magnitude) of a quaternion is:

$$\|q\| = \sqrt{a^2 + b^2 + c^2 + d^2}$$

Quaternions with a norm of 1 are called **unit quaternions** and are particularly useful in representing rotations.

4. **Quaternions and Rotations in 3D Space**

- **Rotation Representation**: A unit quaternion $q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(xi + yj + zk)$, where θ is the angle of rotation and (x, y, z) is the normalized axis of rotation, represents a 3D rotation.
- **Rotation of a Vector**: To rotate a vector $\vec{v}$ in 3D, it is embedded into a pure quaternion (with zero real part), and then the rotation is performed using:

$$\vec{v}' = q\vec{v}q^*$$

This formulation avoids the gimbal lock problem that can occur with Euler angles.

5. **Quaternions in Physics**

- **Quantum Mechanics**: Quaternions provide an alternative way to represent spinors and SU(2) symmetry, linking to Pauli matrices in describing spin states.
- **Special Relativity**: Quaternions are sometimes used to describe Lorentz transformations and boosts in spacetime, though they are more commonly supplanted by biquaternions or spacetime algebra.
- **Electromagnetism**: Maxwell's equations can be compactly represented using quaternions, which neatly combine electric and magnetic fields.

6. **Properties of Quaternion Groups**

- **Lie Group Representation**: The group of unit quaternions corresponds to the Lie group $SU(2)$, which double-covers the rotation group $SO(3)$. This means each rotation in 3D space can be represented by two unit quaternions.
- **Non-Commutativity**: This property is key in the non-abelian nature of rotations, making quaternions suitable for physical theories that require careful handling of rotational order.

7. **Applications in Computer Graphics and Robotics**

- **3D Graphics and Animations**: Quaternions are used to interpolate rotations (slerp—spherical linear interpolation) smoothly without artifacts.
- **Robotics**: Quaternions provide a compact and computationally efficient method for managing the orientation of robotic arms and drones.

8. **Dual Quaternions**

- Dual quaternions extend quaternions to handle not just rotations but also translations, making them useful for representing rigid body transformations in space.

Summary

Quaternions are invaluable in physics due to their ability to represent rotations compactly, their robustness against gimbal lock, and their suitability for high-dimensional algebraic manipulations. They provide an elegant mathematical framework that finds application across a wide array of physical theories and computational tasks.

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Turnpair 2 of 4 ::

@ 2024-09-25 18:49:22 Wed [Δt 0d 00h 00m 46s]

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Now tell me about octonions

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *ab8dbf2a-7c9f-45b5-b806-c5771c8f7ef8*
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Octonions are an extension of quaternions and represent an eight-dimensional number system. They are part of a sequence of hypercomplex numbers that includes real numbers, complex numbers, and quaternions. Octonions play a role in mathematical physics, particularly in theories of exceptional Lie groups, string theory, and certain algebraic structures. Here's what you need to know about octonions:

1. **Definition of Octonions**

Octonions extend quaternions into an eight-dimensional algebra over the real numbers. An octonion O can be written as:

$$O = a_0 + a_1e_1 + a_2e_2 + a_3e_3 + a_4e_4 + a_5e_5 + a_6e_6 + a_7e_7$$

where:

- $a_0, a_1, \ldots, a_7$ are real numbers.
- $e_1, e_2, \ldots, e_7$ are the imaginary units of the octonions.

2. **Octonion Algebra**

- Octonions are neither commutative nor associative; they are **non-commutative** and **non-associative**. However, they are **alternative**, meaning any subalgebra generated by two elements is associative.

- The multiplication rules for octonions are defined by the Fano plane, a seven-point diagram that encodes the multiplication of the seven imaginary units e_1 through e_7.

Fano Plane Multiplication Rules:

1. Each directed line on the Fano plane corresponds to a cyclic multiplication rule.
2. For example, if e_1, e_2, e_3 lie on a line in a cyclic order, then $e_1e_2 = e_3$, $e_2e_3 = e_1$, and $e_3e_1 = e_2$.
3. Multiplication by reversing the order negates the result: $e_2e_1 = -e_3$.

3. **Octonion Conjugate and Norm**

- The **conjugate** of an octonion $O = a_0 + a_1e_1 + \ldots + a_7e_7$ is:

$$O^* = a_0 - a_1e_1 - a_2e_2 - \ldots - a_7e_7$$

- The **norm** of an octonion is given by:

$$\|O\| = \sqrt{a_0^2 + a_1^2 + \ldots + a_7^2}$$

Similar to quaternions, unit octonions (octonions with norm 1) can represent generalized rotations.

4. **Non-Associativity and Implications**

- **Non-Associativity**: Octonions satisfy the property $(xy)z \neq x(yz)$. Instead, they follow a weaker property known as **alternativity**, meaning expressions involving two repeated elements are associative, such as $(xx)y = x(xy)$.

- Non-associativity is rare in mathematical structures, making octonions unique. It complicates their algebraic handling but also hints at deeper symmetries and structures in higher-dimensional physics.

5. **Applications in Mathematical Physics**

- **Exceptional Lie Groups**: Octonions are closely related to exceptional Lie groups, particularly G_2, which is the automorphism group of the octonions. These groups play roles in higher-dimensional geometry and theoretical physics.
- **String Theory and M-Theory**: Octonions appear in the study of string theory and M-theory, especially in contexts involving E_8 symmetry, one of the largest exceptional Lie groups.
- **Quantum Mechanics and Particle Physics**: While direct physical interpretations are complex, octonions are investigated for modeling certain symmetries in particle physics and gauge theories.

6. **Octonions and Spinors**

- **Spinors and Clifford Algebras**: Octonions are linked to the algebraic structure of spinors, particularly in dimensions higher than those typically covered by quaternions. They play a role in the construction of higher-dimensional Clifford algebras.

7. **Challenges and Unique Features**

- **Non-Commutative and Non-Associative**: The lack of both commutativity and associativity poses significant challenges in computation and interpretation, limiting practical use compared to quaternions. However, this complexity also makes octonions an interesting subject for exploring algebraic structures.
- **Normed Division Algebras**: Octonions are part of the four normed division algebras (real numbers, complex numbers, quaternions, and octonions), forming a rare and complete set. They satisfy the property that the product of two octonions' norms equals the norm of their product.

Summary

Octonions are a fascinating, highly structured, and complex number system that extends beyond the familiar properties of commutative and associative algebra. Their non-associativity and relation to exceptional Lie groups suggest deep connections to theoretical physics, particularly in high-energy, quantum, and string theories. Despite their algebraic complexity, they provide valuable insights into the symmetry and structure of the mathematical landscape underlying physical theories.

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Turnpair 3 of 4 ::

@ 2024-09-25 18:52:17 Wed [Δt 0d 00h 02m 38s]

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Last time we said this:

```
We said this:
Our conversation was this:
Here's the multiplication table for the 7D cross product using the specified
variable substitutions:

e
1
→
τ
e
1

 →τ
e
2
→
x
e
2

 →x
e
3
→
t
x
e
3

 →t
x


e
4
→
y
e
4

```

```
 →y
e
5
→
t
y
e
5

 →t
y


e
6
→
z
e
6

 →z
e
7
→
t
z
e
7

 →t
z


Multiplication Table with Substituted Variables
×
τ
x
t
x
y
t
y
z
t
z
τ
0
t
x
−
```

x
t
y
$-$
y
$-$
t
z
z
x
$-$
t
x
0
τ
z
t
z
$-$
y
$-$
t
y
t
x
x
$-$
τ
0
t
z
$-$
z
t
y
$-$
y
y
$-$
t
y
$-$
z
$-$
t
z
0
τ
x
t

```
x
t
y
y
−
t
z
z
−
τ
0
−
t
x
x
z
t
z
y
−
t
y
−
x
t
x
0
−
τ
t
z
−
z
t
y
y
−
t
x
−
x
τ
0
×
τ
x
t
x
```

```
y
t
y


z
t
z





τ
0
−t
x


x
−t
y


y
t
z


−z


x
t
x


0
−τ
−z
−t
z


y
t
y
```

```
t
x


−x
τ
0
−t
z


z
−t
y


y


y
t
y


z
t
z


0
−τ
−x
−t
x




t
y


−y
t
z


−z
τ
0
```

```
t
x


−x


z
−t
z


−y
t
y


x
−t
x


0
τ


t
z


z
−t
y


−y
t
x


x
−τ
0




Explanation:
The table shows the cross product of the substituted vectors, illustrating
how each pair multiplies.
```

Each cell
(
row
×
column
)
(row×column) represents the resulting vector from the cross product of the vectors indicated by the corresponding row and column.
Antisymmetry (
τ
×
x
=
−
(
x
×
τ
)
τ×x=−(x×τ)) and orthogonality properties of the 7D cross product are preserved under these substitutions.
Let me know if you need any further adjustments or explanations!

You said:
how would you interpret the results of the terms where t_i is crossed with i, if the t_i coordinates represent the experience of an observer treating i as their 'self' location?
ChatGPT said:
ChatGPT
If we interpret the coordinates
t
i
t
i

as representing the experience of an observer who treats the corresponding spatial coordinate
i
i as their "self" location, then the cross products
t
i
×
i
t
i

×i have a meaningful interpretation in terms of how an observer's experience or perspective interacts with their self-location in higher-dimensional space. Here's a detailed interpretation of these terms:

Interpretation of
t
i
×
i
t
i

×i:
Terms of Interest:

Specifically, we focus on the terms:
t
x
×
x
t
x

×x
t
y
×
y
t
y

×y
t
z
×
z
t
z

×z
Conceptual Context:

Each
t
i
t
i

represents the experience or perception of an observer located at

i
i. For instance:
t
x
t
x

is the experience related to being at the location
x
x.
t
y
t
y

is the experience related to being at the location
y
y.
t
z
t
z

is the experience related to being at the location
z
z.
The cross product
t
i
×
i
t
i

×i captures how an observer's experience (or time-like dimension) interacts with their spatial self-location.
Result of the Cross Products:

From the table:
t
x
×
x
=
τ
t
x

×x=τ
t

y
×
y
=
τ
t
y

×y=τ
t
z
×
z
=
τ
t
z

×z=τ
Interpretation of the Results:

Each of these results (
t
i
×
i
=
τ
t
i

×i=τ) indicates that when the observer’s experience dimension
t
i
t
i

is crossed with their own spatial location
i
i, the result is a time-like or temporal component, represented by
τ
τ.
This suggests a deep connection between the experience of an observer and their intrinsic time perception:
Temporal Emergence: The interaction of an observer’s experience with their self-location yields a time-like result, implying that subjective time (
τ
τ) is fundamentally linked to how the observer interacts with their spatial position.
Self-Time Connection: The results imply that an observer's sense of time (

```
τ
τ) is not an external parameter but rather emerges directly from the
interaction between where they perceive themselves to be and their experience
of that location.
Non-Detachment of Time: Time (
τ
τ) is not detached from space; rather, it is generated through the personal
interaction of space and self-experience. This reinforces a holistic view
where the observer's internal sense of time is inherently tied to their
spatial context.
Broader Implications:
Subjectivity of Time: The emergence of
τ
τ from these specific cross products highlights the subjective nature of
time, reflecting that time may be a constructed dimension emerging from how
individuals experience their position in space.
Dimension Coupling: These results suggest that in higher-dimensional
frameworks, dimensions representing "experience" are not just passively
coupled with space but actively produce temporal-like outcomes that might
explain the flow of time as perceived by the observer.
In summary, the cross products
t
i
×
i
=
τ
t
i

 ×i=τ can be seen as expressing a fundamental relationship between an
observer's self-location and their experience, manifesting as a personal
temporal dimension, reinforcing a deeply interconnected nature of space,
self, and time.

END HISTORY:

NOTES:
Call the R7 space 'timespace' or 'experience space'. the x, y, z, components
are position space. (tau, x, y, z,) is spacetime. the tx, ty, and tz
components are temporal space. In the limit of only considering an object's
experience from its individual point of view, that is a 4D temporal space
(tau, tx, ty, tz). In the limit of only considering the same object's
experience from an outside observer's point of view, that is the usual 4D
spacetime (tau, x, y, z). The ordering for the R7 tuple is very important.
Also, tau is the same time dimension as in minkowski spacetime.
```

I want you to attempt to apply quaternions to this R7 space in the same way that minkowski applied quaternions to R3 space.

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
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Applying quaternions to the 7D space (R7), which we refer to as "timespace" or "experience space," follows a similar conceptual approach as Minkowski's application of quaternions to 3D space. However, in the case of timespace, we have a richer structure because of the additional temporal components (t_x, t_y, t_z) alongside the conventional spatial coordinates (x, y, z) and the shared time dimension τ. Here's an attempt to extend quaternions into this context:

1. **Defining Quaternions in Timespace (R7)**

Timespace Structure:

- Timespace consists of the coordinates $(\tau, x, t_x, y, t_y, z, t_z)$.
- The coordinates (x, y, z) represent spatial dimensions.
- The coordinates (t_x, t_y, t_z) represent temporal experiences tied to the corresponding spatial coordinates.
- τ is the universal time dimension shared between timespace and spacetime.

2. **Applying Quaternions to Timespace**

Quaternions in Traditional R3 Space:

- In 3D, a quaternion q represents a combination of one real part (scalar) and three imaginary parts (vector): $q = a + bi + cj + dk$.
- Quaternions encode rotations and reflections, enabling transformations within 3D space.

Quaternions in Timespace (R7):

- We aim to extend quaternions to represent rotations, reflections, and possibly more complex interactions in timespace.

- A quaternion-like element Q in timespace could be structured as:

$$Q = a + bi + cj + dk + el + fm + gn + ho$$

where:

- a is the scalar part.
- (i, j, k, l, m, n, o) are imaginary units representing the different axes in timespace.
- Specifically, we can map these units to timespace coordinates as follows:
    - i corresponds to x.

  - j corresponds to y.
  - k corresponds to z.
  - l corresponds to t_x.
  - m corresponds to t_y.
  - n corresponds to t_z.
  - o corresponds to τ.

3. **Multiplication Rules and Rotations in Timespace**

Extension of Quaternion Multiplication:

- Just as quaternion multiplication is defined by specific rules (non-commutative and associative), timespace quaternions would follow multiplication rules extended from the standard quaternion algebra.
- For the extended imaginary units, we would establish a multiplication table preserving the non-commutative and alternative properties, akin to how the Fano plane structures octonion multiplication but adapted for timespace.

Rotation Representation in Timespace:

- A rotation in timespace would involve transforming the spatial and temporal coordinates. For example, a unit quaternion Q that represents a rotation could take the form:

$$Q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(xi + t_x l + yj + t_y m + zk + t_z n)$$

- This would rotate both spatial and temporal components simultaneously, reflecting how an observer's perception and experience interact with the traditional spatial dimensions.

4. **Interpretation of Quaternion Action on Timespace**

Spatial vs. Temporal Interactions:

- In traditional quaternions, the multiplication rules $i^2 = j^2 = k^2 = ijk = -1$ lead to vector transformations preserving the norm.
- In timespace, similar interactions between (i, j, k) and (l, m, n) could reflect the non-commutative nature of how spatial and temporal experiences interweave, producing transformations that encode both spatial rotations and shifts in temporal perception.

Time-Emergent Behaviors:

- Terms like $l \times i$, $m \times j$, and $n \times k$ could reflect how temporal perceptions t_x, t_y, t_z interact with spatial coordinates x, y, z, generating emergent time-like effects represented by τ.
- This extension suggests that the interactions within timespace aren't merely spatial but also intrinsically linked to how time emerges from the observer's engagement with the space around them.

5. **Potential Applications and Implications**

Unified Rotations:

- The quaternion extension provides a unified way to describe rotations and transformations that include not just the physical space but also the temporal experience dimensions, offering a more holistic framework.

Higher-Dimensional Physics:

- This approach aligns with the idea that our perception of time may be deeply rooted in our interactions with space, suggesting possible applications in areas exploring time's subjectivity or higher-dimensional physical models, like generalizations of relativity.

Symmetry and Dynamics:

- These extended quaternions could be used to model complex symmetries in physics where time is not an external parameter but an emergent property from deeper dimensional interactions.

Summary

Applying quaternions to timespace (R7) involves expanding their traditional role to encompass both spatial and temporal dimensions, reflecting a complex interplay of experience and location. This quaternionic approach enables a richer representation of rotations and transformations that naturally merge an observer's spatial and temporal perspectives, aligning with Minkowski's conceptual leap from R3 to R4 in relativity but extended into a more subjective, experience-oriented framework.

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Turnpair 4 of 4 ::

@ 2024-09-25 19:00:57 Wed [Δt 0d 00h 08m 14s]

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

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But I want to mimic the same R3 to R4 leap, but R7 to R8, where the extra dimension is similarly the time component that is a complex value to produce the metric (+,-,-,-,-,-,-).

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *64407058-3efc-4152-80fd-d0c8107eeafd*
> ***parent:*** *aaa238bb-e380-44a5-aacc-db7744db311f*
> ***children:***

To extend the transformation from R7 (timespace) to R8 in a manner analogous to Minkowski's leap from R3 to R4, we need to construct a higher-dimensional space where the additional dimension plays a similar role to time in Minkowski spacetime. This new dimension will be treated in a way that aligns with your intention to produce the metric signature $(+,-,-,-,-,-,-,-)$, integrating complex values as part of the extension.

R7 to R8 Leap: Defining the Extra Dimension

1. **Timespace (R7) Recap:**
   - The coordinates of R7 are $(\tau, x, t_x, y, t_y, z, t_z)$.
   - τ is a shared time dimension, analogous to the time component in spacetime.
   - (x, y, z) are spatial coordinates.
   - (t_x, t_y, t_z) are temporal components tied to spatial directions, representing an observer's experience of time.

2. **Extending to R8:**
   - The goal is to add a new dimension ϕ that serves as an overall temporal component, analogous to how time τ integrates into spacetime.
   - This dimension will be complex-valued to facilitate the desired metric signature: $(+,-,-,-,-,-,-,-)$.

3. **New Space Definition (R8):**
   - We define the extended space $\mathbb{R}^8$ as $(\phi, \tau, x, t_x, y, t_y, z, t_z)$.
   - ϕ serves as a complex time dimension, extending the notion of time in a way that captures both standard and subjective experiences of time.
   - ϕ is a complex value, represented as $\phi = \tau' + i\psi$, where:
     - τ' can be thought of as a new real time-like component.
     - ψ introduces an imaginary component, adding a richer, possibly oscillatory or phase-related aspect to time.

Applying the Metric $(+,-,-,-,-,-,-,-)$

1. **Metric Construction:**
   - The desired metric signature is $(+,-,-,-,-,-,-,-)$, indicating:
     - ϕ contributes a positive component.
     - All spatial and temporal coordinates contribute negative components.

2. **Metric Tensor Representation:**

- The metric tensor $g_{\mu\nu}$ for R8 can be structured as:

$$g_{\mu\nu} = \begin{bmatrix} +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

3. **Interpretation of ϕ:**

   - The complex dimension ϕ is treated as contributing to time-like behavior but with additional dynamics, reflecting a richer set of interactions between time and experience.
   - The imaginary part ψ suggests phenomena such as cyclical time, phase shifts, or complex time-dependent transformations.

Connecting Quaternions to R8:

1. **Quaternion Representation:**

   - Quaternions, as used in standard spacetime, could now be extended to work within this new R8 setting.
   - A generalized quaternion Q for R8 could involve components that interact between ϕ, τ, and spatial-temporal vectors, generating new types of rotations or transformations.

2. **Possible Applications of Complex Time:**

   - The complex time dimension ϕ allows for modeling effects such as time dilation, oscillations in time perception, or even quantum-like time behaviors.

3. **Implications of the Metric:**

   - The metric $(+, -, -, -, -, -, -, -)$ captures how ϕ introduces a distinct, positively contributing temporal direction.
   - This mirrors the mathematical extension from R3 to R4 but now adapts the concept to integrate both physical and experiential aspects of time.

Summary

This R7 to R8 extension aligns conceptually with Minkowski's transition from R3 to R4 but operates within a more complex landscape, reflecting not just physical time but an enhanced temporal reality influenced by the observer's experience. By introducing ϕ as a complex time dimension, the construction preserves the structure of timespace while embedding a richer dynamic, capturing both familiar and novel temporal phenomena under the new metric signature.

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